MyMavenWebApp

Step 1: Generate Maven Web App Archetype

Navigate to the project directory and execute the mvn archetype:generate command. Use the maven-archetype-webapp artifact to create a template project with the groupId set to com.example and artifactId as MyMavenWebApp001.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~$ cd Devops/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -Da
rchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 24 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shravan-hegde/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/Devops/MyMavenWebApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.885 s
[INFO] Finished at: 2026-04-08T08:26:51+05:30
[INFO] ------------------------------------------------------------------------
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$
```

Step 2: Verify Initial Web App Structure

Enter the newly created project folder and use the tree command. This confirms the standard Maven web layout, including the src/main/webapp directory, the WEB-INF folder, and the initial index.jsp file.

```
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shravan-hegde/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp001
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/Devops/MyMavenWebApp001
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.853 s
[INFO] Finished at: 2026-04-08T08:32:29+05:30
[INFO] ------------------------------------------------------------------------
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops$ cd MyMavenWebApp001/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenWebApp001$ tree
.
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

6 directories, 3 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenWebApp001$
```

Step 3 : Configure Project Dependencies

Open the pom.xml file using a text editor like gedit. Add the necessary dependencies for the project, such as the JUnit library for testing and the Java Servlet API for web functionality, ensuring the packaging type is set to war.

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MymavenWebApp01</artifactId>
  <packaging>war</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MymavenWebApp01 Maven Webapp</name>
  <url>http://maven.apache.org</url>


  <dependencies>


    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>

    <!-- Servlet API for Java EE -->
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>javax.servlet-api</artifactId>
            <version>4.0.1</version>
            <scope>provided</scope>
        </dependency>


  </dependencies>


  <build>
  <plugins>
  <plugin>
```

Step 4 : Execute Clean Install and Build

Run the mvn clean install command to initiate the full build lifecycle.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops/MyMavenWebApp001$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MymavenWebApp01 >---------------------
[INFO] Building MymavenWebApp01 Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MymavenWebApp01 ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MymavenWebApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MymavenWebApp01 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MymavenWebApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/devops/MyMavenWebApp001/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MymavenWebApp01 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MymavenWebApp01 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MymavenWebApp01 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MymavenWebApp01] in [/home/shravan-hegde/devops/MyMavenWebApp001/target/MymavenWebApp01]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/shravan-hegde/devops/MyMavenWebApp001/src/main/webapp]
[INFO] Building war: /home/shravan-hegde/devops/MyMavenWebApp001/target/MymavenWebApp01.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MymavenWebApp01 ---
[INFO] Installing /home/shravan-hegde/devops/MyMavenWebApp001/target/MymavenWebApp01.war to /home/shravan-hegde/.m2/repository/com/examp
le/MymavenWebApp01/1.0-SNAPSHOT/MymavenWebApp01-1.0-SNAPSHOT.war
[INFO] Installing /home/shravan-hegde/devops/MyMavenWebApp001/pom.xml to /home/shravan-hegde/.m2/repository/com/example/MymavenWebApp01/
```

Step 5 : Inspect Final Build Artifacts

After the build completes, use the tree command again to inspect the target folder. Verify that the MymavenWebApp01.war file has been generated along with the exploded webapp directory structure.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops/MyMavenWebApp001$ tree
.
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MymavenWebApp01
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MymavenWebApp01.war

13 directories, 7 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops/MyMavenWebApp001$
```

Step 6 : Verify Web Application Interface

Open a web browser and navigate to the project's local path to access the index.jsp file. A successful deployment is confirmed when the browser correctly renders the "Hello World!" message from the web application.